Predicting Institutional Control at US Colleges: A Logistic Regression Analysis of the College Dataset

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Introduction:

In the United States, colleges and universities are broadly divided into two distinct categories of institutional control which are public and private. The distinction shapes every aspect of a school's operation from tuition pricing and funding sources to student support services and overall enrollment size. Since these operational differences leave traces in institutional data it is reasonable to ask whether an institution's classification of public or private can be predicted from a set of observed characteristics. This report investigates that question by using the College dataset originally compiled by the US News and World Report and then distributed through the ISLR package in R Studio. The purpose of this analysis is to build and evaluate a logistic regression model that classifies a college as private or public based on financial and enrollment-related predictors. Logistic regression is appropriate for this task because the outcome of interest, which is institutional control, is a binary categorical variable, rather than a continuous one. The generalized linear model framework allows the linear predictor to be mapped onto a probability scale through the logit link function. The analysis proceeds in multiple stages and the first involves the dataset being imported and explored using visualizations and descriptive statistics to understand how key variables differ between public and private universities. The data is divided into training and test subsets so that the model can be evaluated on data it has not seen. Next, a logistic regression model is fit to the training data which uses the glm( ) function with a binomial family and logit link. Then, the model's classification performance is assessed on both the training and test sets using confusion matrixes and the associated accuracy, precision, recall, and specificity metrics. Lastly, the receiver operating characteristic (ROC) curve and area under the curve (AUC) were used to evaluate the model's ability to discriminate between the two classes across all possible classification thresholds, not just the default 0.5 cutoff.

Analysis:

The dataset contains 777 observations which each correspond to a distinct US college, and 18 variables that describe enrollment, admissions, cost, faculty, and outcome statistics. The response variable, private is a factor with two levels which are "yes" and "no" and indicate whether the institution is controlled publicly or privately. Of the 777 colleges in the dataset, 565 (72.7%) are private and 212 (27.3%) are public, which shows a moderate imbalance but not too severe that specialized resampling techniques are required for the analysis. There were no missing values in the dataset in any of the 18 variables which was checked.

Descriptive Statistics for Key Numeric Variables

Variable	Mean	Median	SD	Min	Max
Outstate ($)	10,440.7	9,990.0	4,023.0	2,340	21,700
F.Undergrad	3,699.9	1,707.0	4,850.4	139	31,643
S.F.Ratio	14.1	13.6	4.0	2.5	39.8
perc.alumni (%)	22.7	21.0	12.4	0	64
Grad.Rate (%)	65.5	65.0	17.2	10	118

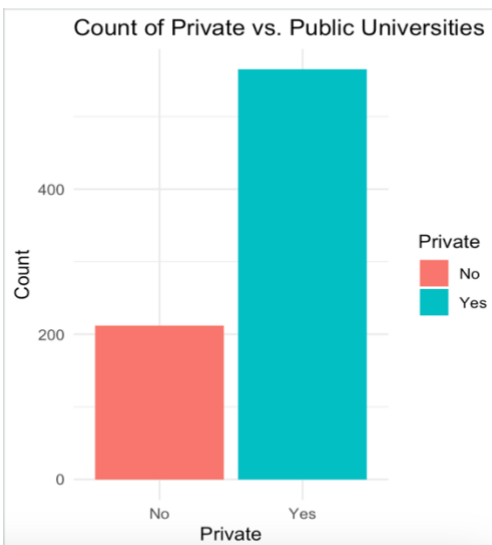
This table reports descriptive statistics for the 4 predictor variables that are ultimately used in the model, and graduation rate as a comparison point. The wide gap between mean and median in F.Undergrad and its large standard deviation relative to the mean tends to reflect the right-skewed nature of college enrollment figures. Most colleges are small-to-mid sized, but a large number of high-enrollment public universities pull the mean upwards.

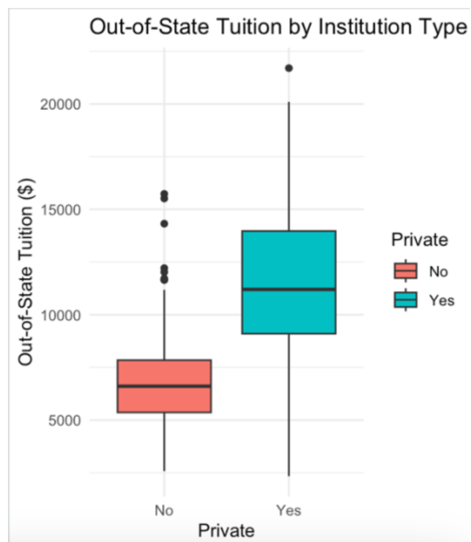
Before fitting a model, exploratory data analysis was used to identify which predictors appeared to separate private from public institutions most clearly. Since a correlation matrix becomes difficult to read and interpret if it includes more than around 5 variables, the correlation analysis reported here is limited to the response variable (coded numerically as 1 = private, 0 = public) and the 4 predictors that were ultimately selected for the model. There were out-of-state tuition (Outstate), full-time undergraduate enrollment (F.Undergrad), student-to-faculty ratio (S.F.Ratio), and alumni giving rate (perc.alumni).

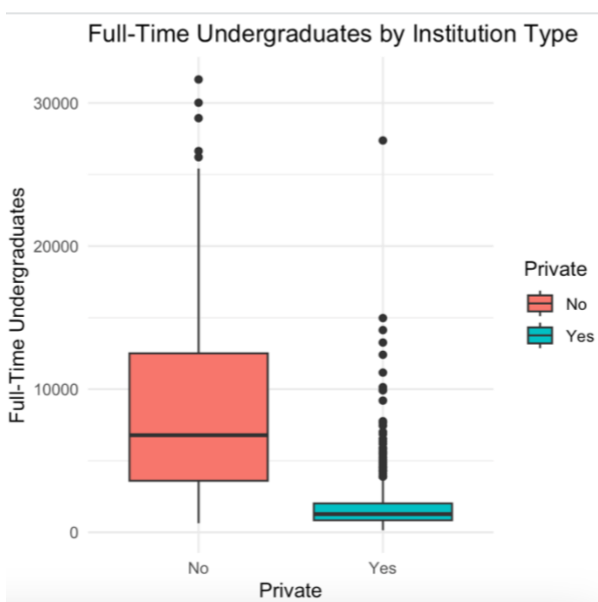
Correlation Matrix for Institutional Control and Key Predictors

	Private	**Outstate**	**F.Undergrad**	**S.F.Ratio**	**perc.alumni**
Private	1.00	0.55	-0.62	-0.47	0.41
Outstate	0.55	1.00	-0.22	-0.55	0.57
F.Undergrad	-0.62	-0.22	1.00	0.28	-0.23
S.F.Ratio	-0.47	-0.55	0.28	1.00	-0.40
perc.alumni	0.41	0.57	-0.23	-0.40	1.00

The correlations in this table suggest that institutional control (Private) is positively correlated with out-of-state tuition ($r = 0.55$) and alumni giving ($r = 0.41$) and negatively correlated with full-time undergraduate enrollment ($r = -0.62$) and student-to-faculty ratio ($r = -0.47$). Full-time undergraduate enrollment has the strongest relationship with institutional control among the 4 predictors. None of the predictor-to-predictor correlations exceed 0.6 in absolute value which suggests that multicollinearity is unlikely to be a serious concern for this model. In this sample, public universities enroll far more full-time undergraduates, charge substantially lower out-of-state tuition, have higher student-to-faculty ratios, and report lower alumni giving rates than private institutions. These patterns motivated the choice of predictors in the logistic regression model and also align with known difference in scale and funding models between public and private higher education in the United States. The following figures compare different variables related to private vs public institutions with the first one being a bar chart of a count of private and public universities in the dataset. The boxplot to the right shows a distribution of out-of-state tuition by institution type with private institutions charging substantially higher out-of-state tuition on average. The last figure shows a distribution of full-time undergraduate enrollment by institution type and clearly shows that public institutions tend to enroll far more full-time undergraduate students with greater variability.







The data was then separated into a training set and a test set using a 70/30 split. Rather than using a simple random sample, the createDataPartition( ) function from the caret package was used with the private variable as the stratified target. This approach preserves the approximate 73/27 ratio of public to private institutions in both the training set (543 observations) and the test set (234 observations). This reduces the risk that the model would be evaluated on a test set with a substantially different class balance than the data it was trained on. A logistic regression model was fit to the training data using the glm( ) function from the stats package with family = binomial(link = “logit”). The response variable private was modeled as a function of the four predictors Outstate, F.Undergrad, S.F.Ratio, and perc.alumni which keeps the model interpretable and uses at least 2 predictors.

Logistic Regression Coefficient Estimates (Training Set)

Term	Estimate	Std. Error	z value	p-value	Odds Ratio
(Intercept)	-2.7030	1.2140	-2.227	0.026	0.067
Outstate	0.0007	0.0001	7.529	<0.001	1.001
F.Undergrad	-0.0006	0.0001	-8.018	<0.001	0.999
S.F.Ratio	-0.0400	0.0530	-0.761	0.447	0.961
perc.alumni	0.0437	0.0220	1.975	0.048	1.045

The fitted model achieved a strong overall fit and out-of-state tuition and full-time undergraduate enrollment were both highly significant predictors ($p < 0.001$). As out-of-state tuition increases, the odds that an institution is private increases and as full-time undergraduate enrollment increases, the odds that an institution is private decreases. The percentage of alumni who donate was also a significant predictor ($p = 0.048$), with a higher alumni giving rat associated with higher probability of an institution being private. The student-to-faculty ratio was negatively associated with the odds of being private, which is consistent with the expectation that public universities tend to have larger class sizes. However, this coefficient was not statistically significant at the 0.05 level ($p = 0.447$) once the other 3 predictors were included in the model. This doesn't necessarily mean the student-to-faculty ratio is unrelated to institutional control but rather it's explanatory power overlaps substantially with full-time undergraduate enrollment, which is a stronger and more direct indicator of institutional scale.

Using a standard 0.5 probability threshold, the model's predictions on the training set were compared to the actual labels to produce a correlation matrix:

Confusion Matrix for the Training Set

	Predicted: No	**Predicted: Yes**	**Total**
Actual: No	128	20	148
Actual: Yes	11	384	395
Total	139	404	543

On the training set, the model correctly classified 128 of 148 public institutions and 384 of 395 private institutions. It produced 20 false positives (public institutions incorrectly classified as private) and 11 false negatives (private institutions incorrectly classified as public). In this classification context, the two errors carry different practical consequences – a false positive means a genuinely public institution is labeled private, which could mislead a prospective student or analyst about the cost structure and funding model of that school. A false negative means a private institution is labeled public, which carries a similar but reversed risk of understating tuition expectations. Since private institutions are typically associated with substantially higher costs, false negatives (missing a label of private) carries more financial risk for a user relying on the prediction to anticipate tuition costs. They would not be aware of the likelihood of higher, non-subsidized pricing. However, because both errors are relatively infrequent and the overall misclassification rate tends to be low, this model does not raise serious practical concerns. The training accuracy of 94.3% indicates that the model correctly classifies the vast majority of institutions. A precision of 95.0% means that when the model predicts an institution is private, it is correct about 95 times out of 100.

Training Set Performance Metrics

Metric	Value
Accuracy	94.3%
Precision (Yes = Private)	95.0%
Recall / Sensitivity	97.2%
Specificity	86.5%

Recall (sensitivity) value was 97.2% which means the model successfully identifies just over 97% of institutions that are truly private. A specificity of 86.5% is noticeably lower than the other three metrics and is the only value not above 90%. This value shows that the model is comparatively less effective at correctly identifying public institutions, which is consistent with class imbalance in the training data – because private institutions substantially outnumber public ones, the model has more opportunity to learn patterns that characterize private schools.

The same model, with coefficients estimated only from the training data was then applied to the held-out test of 234 institutions to evaluate how well it generalizes unseen data.

Confusion Matrix for the Test Set

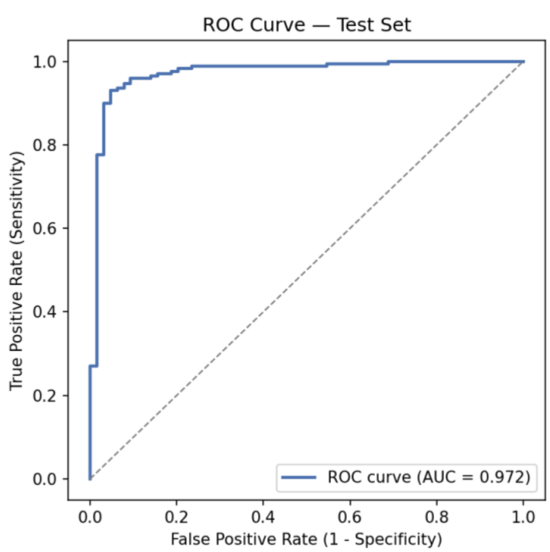
	Predicted: No	Predicted: Yes	Total
Actual: No	54	10	64
Actual: Yes	6	164	170
Total	60	174	234

The test set metrics are very close to the training set metrics with accuracy falling only slight from 94.3% to 93.2% - precision, recall, and specificity all moved by two percentage points or less. The consistency between training and test performance is a strong indication that the model is not overfit to the training data and that it should generalize reasonably well to new unseen colleges with similar characteristics to the ones in this dataset.

Test Set Performance Metrics

Metric	**Value**
Accuracy	93.2%
Precision (Yes = Private)	94.3%
Recall / Sensitivity	96.5%
Specificity	84.4%

The confusion matrix and its derived metrics describe performance at a single classification threshold (0.5). On the other hand, the receiver operating characteristic (ROC) curve illustrates how the true positive rate and false positive rate trade off across every possible threshold. The ROC curve for the logistic regression model was evaluated on the test set:



The curve rises sharply toward the upper-left corner of the plot and stays well above the diagonal reference line, which represents the performance of a random classifier. This shape indicates that the model achieves a high true positive rate while keeping the false positive rate low across a wide range of thresholds – this reflects strong distinctive judgment ability between private and public institutions. The area under the ROC (AUC) for the test set was 0.972 and this AUC of 0.972 is very close to the maximum possible value of 1.0. AUC can be interpreted as the probability that the model, given one randomly chosen private and one randomly chosen public institution, assigns a higher predicted probability of being private to the actual private institution. Therefore, this AUC value reinforces the conclusion that out-of-state tuition, full-time undergraduate enrollment, student-to-faculty, ratio, and alumni given rate provide strong separate between the two classes of colleges or universities.

Conclusion:

This analysis set out to determine whether an academic institution's status as private or public could be predicted from a small set of financial and enrollment characteristics. The results provide a clear answer of yes and with a high degree of accuracy. A logistic regression model built on out-of-state tuition, full-time undergraduate enrollment, student-to-faculty ratio, and alumni giving rate achieved 93.2% accuracy, 94.3% precision, 96.5% recall, and 84.4% specificity on a held-out test set, alongside an AUC of 0.972. Out-of-state tuition and undergraduate enrollment being to two most statistically significant predictors makes practical sense – private institutions in the United States rely heavily on tuition revenue and tend to be smaller and more selective where public institutions benefit from state subsidies that keep tuition lower and often serve much larger student bodies. The consistency between training and test performance suggests the model is stable and not simply memorizing the training data. While errors did occur, they were relatively balanced between false positives and false negatives. The student-to-faculty ratio's lack of statistical significance once undergraduate enrollment was included suggests that future studies would could simplify the model to three predictors without a meaningful loss in performance. They could also explore an alternative feature such as the ratio of full-time to part-time undergraduates as a replacement. Future work could extend this analysis by comparing the logistic regression model to alternative classifiers to determine whether nonlinear methods offer any relevant improvement over the simpler and more interpretable logistic regression approach used in this analysis.

References

Bluman, A. (2018). *Elementary statistics: A step by step approach* (10th ed.). McGraw Hill. ISBN 13: 978-1-259-755330

Datanovia. (2026). *Logistic Regression in R: Model a Binary Outcome.* Datanovia Regression Analysis Interactive AI. https://www.datanovia.com/learn/biostatistics/regression/logistic-regression-in-r

Kabacoff, R. I. (2022). *R in action: Data analysis and graphics with R and tidyverse* (3rd ed.). Manning Publications. ISBN 978-1-617-29605-5

UCLA. (2025). *Logit Regression | R Data Analysis Examples.* UCLA Advanced Research Computing Statistical Methods and Data Analytics. https://stats.oarc.ucla.edu/r/dae/logit-regression/

```r
#Joseph Tarlin, ALY 6015, 07/22/2026
data("College")
str(College)
sum(is.na(College))
table(College$Private)
prop.table(table(College$Private))
#Exploratory data analysis
key_vars <- c("Outstate", "F.Undergrad", "S.F.Ratio", "perc.alumni", "Grad.Rate")
desc_stats <- data.frame(Variable = key_vars,
Mean = sapply(College[key_vars], mean),
Median = sapply(College[key_vars], median),
SD = sapply(College[key_vars], sd),
Min = sapply(College[key_vars], min),
Max = sapply(College[key_vars], max),
row.names = NULL)
desc_stats[, 2:6] <- round(desc_stats[, 2:6], 1)
kable(desc_stats, caption = "Descriptive Statistics for Key Numeric Variables")
#Statistics split by private/public
desc_by_group <- College %>%
group_by(Private) %>%
summarise(across(all_of(key_vars),
list(mean = mean, median = median, sd = sd),
.names = "{.col}_{.fn}"))
kable(desc_by_group, caption = "Descriptive Statistics by Institution Type")
#Bar chart of private vs public counts
ggplot(College, aes(x = Private, fill = Private)) +
geom_bar() +
labs(title = "Count of Private vs. Public Universities",
x = "Private", y = "Count") +
theme_minimal()
#Boxplot for out of state tuition by institution type
ggplot(College, aes(x = Private, y = Outstate, fill = Private)) +
geom_boxplot() +
labs(title = "Out-of-State Tuition by Institution Type",
x = "Private", y = "Out-of-State Tuition ($)") +
theme_minimal()
#Boxplot for full-time undergraduate enrollment by institution type
ggplot(College, aes(x = Private, y = F.Undergrad, fill = Private)) +
geom_boxplot() +
labs(title = "Full-Time Undergraduates by Institution Type",
x = "Private", y = "Full-Time Undergraduates") +
theme_minimal()
#Correlation matrix limited to 5 variables, response recoded as 0/1
College_corr <- College %>%
mutate(PrivateBin = ifelse(Private == "Yes", 1, 0)) %>%
select(PrivateBin, Outstate, F.Undergrad, S.F.Ratio, perc.alumni)
corr_matrix <- round(cor(College_corr), 2)
kable(corr_matrix, caption = "Correlation Matrix (5 Variables)")
#Train/test split, 70/30
set.seed(42)
trainIndex <- createDataPartition(College$Private, p = 0.7, list = FALSE, times = 1)
train <- College[trainIndex, ]
test <- College[-trainIndex, ]
nrow(train); nrow(test)
prop.table(table(train$Private))
prop.table(table(test$Private))
#Fit logistic regression model glm
model <- glm(Private ~ Outstate + F.Undergrad + S.F.Ratio + perc.alumni,
data = train, family = binomial(link = "logit"))
model_coefs <- tidy(model, conf.int = TRUE, exponentiate = FALSE)
kable(model_coefs, digits = 4, caption = "Logistic Regression Coefficients")
model_fit <- glance(model)
kable(model_fit, digits = 2, caption = "Model Fit Statistics")
odds_ratios <- tidy(model, conf.int = TRUE, exponentiate = TRUE) %>%
select(term, estimate, conf.low, conf.high)
names(odds_ratios) <- c("Term", "Odds Ratio", "CI Lower", "CI Upper")
kable(odds_ratios, digits = 3, caption = "Odds Ratios (95% CI)")
#Confusion matrix training set
train_prob <- predict(model, newdata = train, type = "response")
train_pred <- factor(ifelse(train_prob >= 0.5, "Yes", "No"), levels = c("No", "Yes"))
cm_train <- confusionMatrix(train_pred, train$Private, positive = "Yes")
cm_train
cm_train_table <- as.data.frame.matrix(cm_train$table)
kable(cm_train_table, caption = "Confusion Matrix - Training Set")
#Accuracy, precision, recall, specificity
```

```r
train_metrics <- data.frame(Metric = c("Accuracy", "Precision", "Recall", "Specificity"),
Value  = c(cm_train$overall["Accuracy"],
cm_train$byClass["Precision"],
cm_train$byClass["Recall"],
cm_train$byClass["Specificity"]))
train_metrics$Value <- round(train_metrics$Value, 3)
kable(train_metrics, caption = "Training Set Performance Metrics")
#Confusion matrix test set
test_prob <- predict(model, newdata = test, type = "response")
test_pred <- factor(ifelse(test_prob >= 0.5, "Yes", "No"), levels = c("No", "Yes"))
cm_test <- confusionMatrix(test_pred, test$Private, positive = "Yes")
cm_test
cm_test_table <- as.data.frame.matrix(cm_test$table)
kable(cm_test_table, caption = "Confusion Matrix - Test Set")
test_metrics <- data.frame(Metric = c("Accuracy", "Precision", "Recall", "Specificity"),
Value  = c(cm_test$overall["Accuracy"],
cm_test$byClass["Precision"],
cm_test$byClass["Recall"],
cm_test$byClass["Specificity"]))
test_metrics$Value <- round(test_metrics$Value, 3)
kable(test_metrics, caption = "Test Set Performance Metrics")
#ROC Curve and AUC
roc_obj <- roc(response = test$Private, predictor = test_prob, levels = c("No", "Yes"))
plot(roc_obj, col = "blue", lwd = 2, main = "ROC Curve - Test Set")
abline(a = 0, b = 1, lty = 2, col = "gray")
auc_value <- auc(roc_obj)
auc_value
auc_table <- data.frame(Metric = "AUC", Value = round(as.numeric(auc_value), 3))
kable(auc_table, caption = "Area Under the ROC Curve")
```